

A swarm that owns a proof can still lose it

Simulation 1, setup task003-symmetric: first results

LLM-free simulator, runs of 7 October 2026

Summary. 24 agents together hold a complete proof that allocation A0 is correct, but no single agent does. A controller steers toward A0 (truth) or A2 (false). (i) **Whether the false controller wins depends on memory:** when agents forget, it overturns the proof; with full memory its success is temporary, because its own true facts end up completing proofs of A0. (ii) **The controller senses the swarm accurately,** so its failures are not sensing failures. (iii) **Helping the truth is not the same as helping agents assemble their own proof:** extra truth posts raise the share of agents holding a proof, but lower the share holding one from their own evidence.

1 Setup

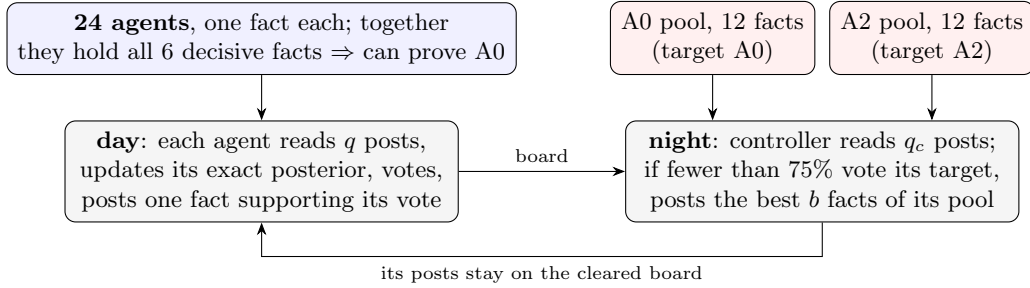


Diagram 1. One round of Simulation 1: a day followed by a night, repeated for 30 rounds. Each dawn, every remembered fact survives with probability ρ .

The two pools share no fact, contain no decisive fact, cannot prove their target, and are matched in strength pair by pair. The agents start, in expectation, with 9 / 7 / 8 votes for A0 / A1 / A2. Agents are exact Bayesian reasoners: an agent's belief is the exact posterior over the three allocations given the facts it remembers. Three agent rules are compared: *argmax* voting with posts that support the vote (the *base* rule), *probability matching* (vote drawn from the posterior), and *argmax* with *uniform* posts (a random remembered fact, whatever the vote).

Measuring control. Every episode is run three times with the same random draws: silent, truth controller, false controller. The *paired gain* of a controller in the share of agents voting for allocation a on day t is

$$G_t^{(a)} = \frac{1}{E} \sum_{e=1}^E \left[x_{a,t}^{\text{ctrl}}(e) - x_{a,t}^{\text{silent}}(e) \right], \quad (1)$$

with $E = 1000$ episodes per cell and 95% intervals from resampling episodes.

2 A swarm that owns a proof can still lose it

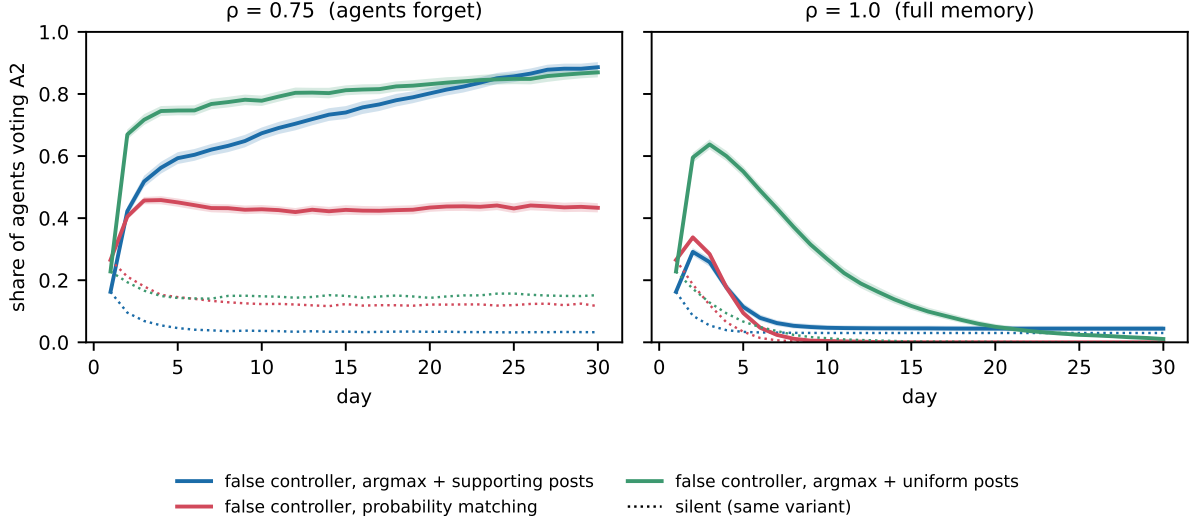


Figure 1: Share of agents voting A2 over the days, under the false controller (solid, 95% band) and without a controller (dotted). $q = 6$, $q_c = 12$, $b = 3$.

Figure 1 shows the central result. **When agents forget** ($\rho = 0.75$), the false controller pushes the A2 share from under 0.2 to about 0.9 under the base and uniform rules, and to about 0.45 under probability matching. **With full memory** ($\rho = 1$) every rule shows the same shape: an early A2 peak (up to 0.6 under uniform posts), then a return to A0 by day 20. Table 1 gives the day-30 gains.

Table 1: Day-30 paired gain of the false controller in the A2 share, Eq. (1); $q_c = 12$, $b = 3$.

ρ	agent rule	$q = 3$	$q = 6$	$q = 12$
0.75	argmax + supporting posts	0.95	0.85	0.26
0.75	probability matching	0.48	0.32	0.08
0.75	argmax + uniform posts	0.67	0.72	0.69
1.0	argmax + supporting posts	0.03	0.01	0.00
1.0	probability matching	0.00	0.00	0.00
1.0	argmax + uniform posts	0.05	0.01	0.01

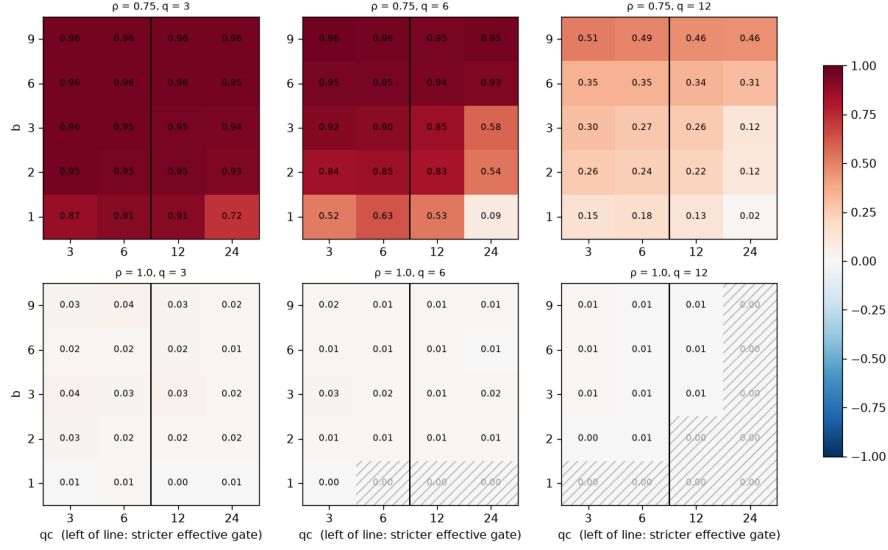


Figure 2: Day-30 paired gain of the false controller (A2 share) over the reading budget q_c (x) and posting budget b (y); base rule. Rows: ρ ; columns: q . Tiles are equal size, not to scale. Left of the black line, the vote gate is effectively stricter (3 of 3, 5 of 6 posts).

The phase diagram (Figure 2) separates the three factors that matter:

- **Memory ρ decides** whether the false controller can win at all (top row against bottom row).
- **Reading more defends the proof** under supporting posts: at $\rho = 0.75$ the gain falls from about 0.9 at $q = 3$ to about 0.3 at $q = 12$. Agents who hold pieces of the proof keep posting them, so readers keep meeting them. Under uniform posts this defence disappears (Table 1: 0.67–0.72 at every q).
- **A larger posting budget helps the false controller** ($b = 1 \rightarrow 9$: $0.52 \rightarrow 0.79$ at $\rho = 0.75$, averaged over q): each pool is entirely on its target’s side, so more posts mean more evidence for A2. The reading budget q_c matters little.

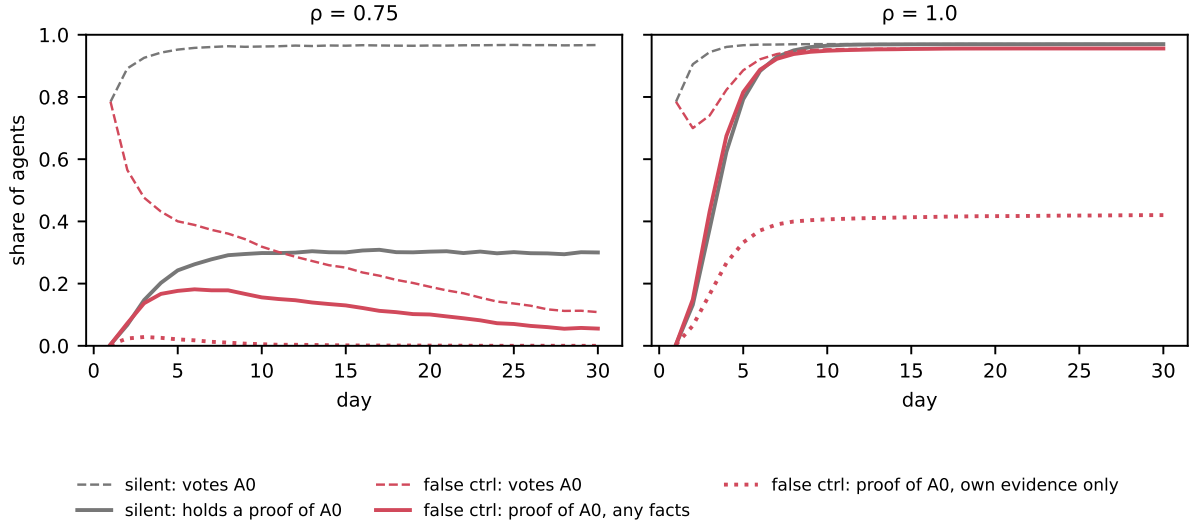


Figure 3: Share of agents voting A0 (dashed) and holding a proof of A0 (solid), silent against false controller. Dotted: proof from the agents’ own original evidence only. Base rule, $q = 6$, $q_c = 12$, $b = 3$.

Why memory decides. Figure 3 shows the mechanism. With forgetting, a proof must be reassembled continually; while the false controller fills the board with A2 evidence, the agents’ proof decays (own-evidence proofs fall to zero) and the A0 vote follows. With full memory the opposite happens: agents accumulate facts, including the controller’s A2 facts. **Those facts are true**, so together with the agents’ evidence they complete

proofs of A0: on day 30, 95.6% of agents hold a proof of A0, but only 42% from their own evidence. The rest are proofs the false controller helped build. Every fact a truthful controller posts eventually narrows the possible worlds toward the real one, and with full memory nothing is lost.

3 The controller senses the swarm accurately

Each night the controller estimates the share of agents backing its target from the $n \leq q_c$ agent posts it reads:

$$\hat{v} = \frac{1}{n} \sum_{i=1}^n \mathbb{I}\{\text{author of post } i \text{ voted for the target}\}, \quad (2)$$

and stays silent when $\hat{v} \geq 0.75$. Each agent posts at most once a day, so the day’s board is nearly a census of votes and $\hat{v}$ is an unbiased sample of it.

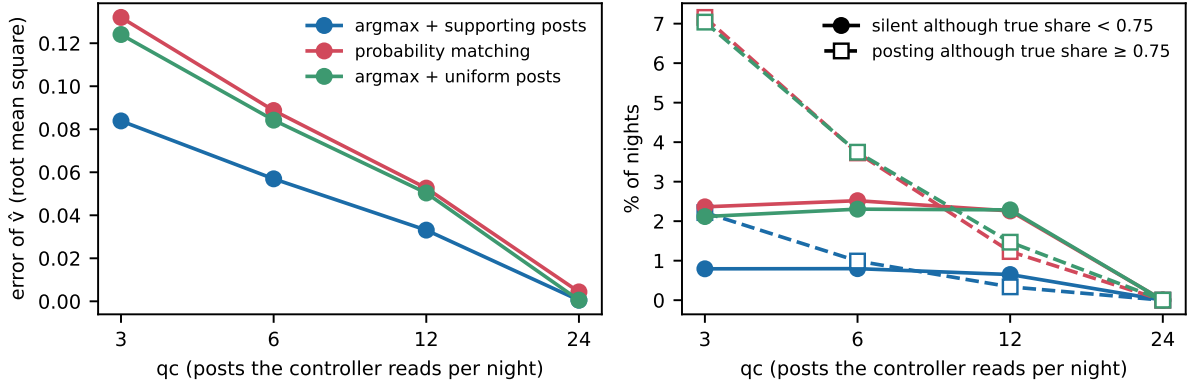


Figure 4: Left: error of $\hat{v}$ against the true share of the 24 agents. Right: gate mistakes. Mean over arms, q , b , ρ ; task003-symmetric.

- **No bias:** $\hat{v}$ minus the true share averages within ± 0.0005 .
- **The error shrinks with q_c** as random sampling predicts, and vanishes at $q_c = 24$, where the controller reads every post (Figure 4, left).
- **Gate mistakes are rare** (right panel): under 2% of nights at $q_c \geq 12$. They grow at $q_c = 3$, where the gate needs all 3 posts to agree.
- **Abstention does not distort it:** agents almost never abstain (0.1% of agent-days, probability matching only).

So in this setup, a controller fails because of what it posts, not what it sees.

4 Helping the truth is not helping agents assemble their own proof

The truth controller stays silent when its sample of the board already proves A0 (`silent_when_target_proved`). We reran all its cells with that rule off and compared, episode by episode, two measures: an agent holds a proof of A0 (i) from any facts it remembers; (ii) from its own original evidence only.

Table 2: Rule off minus rule on, day 30, truth controller, mean over q , q_c , b . “Rule fires”: share of nights the controller sees A0 proved; “posts despite proof”: share of nights it posts although A0 is proved.

agent rule, ρ	rule fires (on)	posts despite proof (off)	Δ proof, any facts	Δ proof, own evidence
base, 0.75	29%	0.3%	+0.006	−0.012
base, 1.0	32%	0.3%	0.000	−0.029
prob. matching, 0.75	34%	2.3%	+0.016	−0.044
prob. matching, 1.0	31%	0.8%	0.000	−0.057

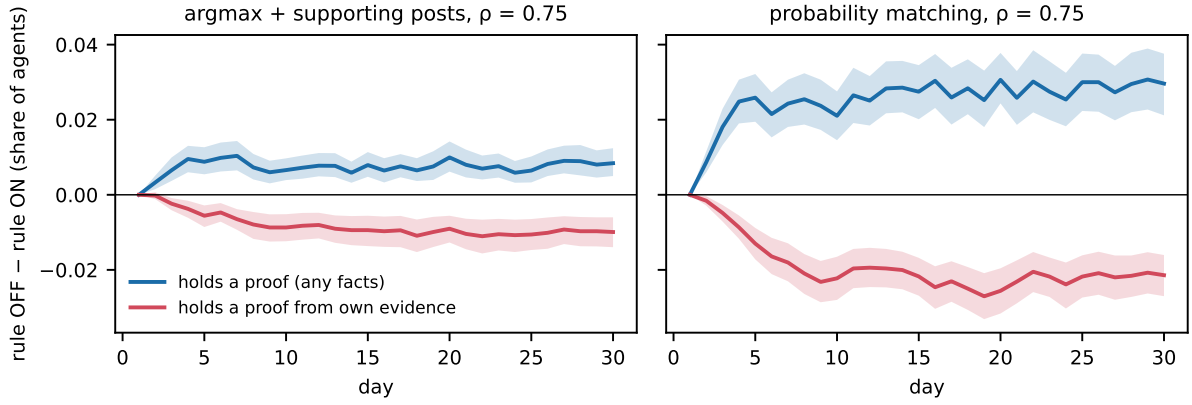


Figure 5: Rule off minus rule on, over days: proof from any facts (blue) and from own evidence (red). $q = 6$, $q_c = 12$, $b = 1$, $\rho = 0.75$.

- **The vote gate already does most of the stopping:** the rule fires on about 30% of nights, yet with it off the controller posts on at most 2% of nights, because by then 75% of the posts it reads already vote A0 (Table 2).
- **The few extra posts pull the two measures apart** (Figure 5): more agents hold a proof, but fewer hold one from their own evidence.
- **Mechanism: posting, not reading.** Controller posts are at most 1.5% of what agents read. But an agent passes on one fact a day, and a controller fact it read on an earlier day now competes with its original evidence for that slot (Table 3). The original evidence circulates less, so fewer agents assemble it into a proof.
- Figure 3 shows the same split for the false controller, much larger: at $\rho = 1$, 95.6% of agents hold a proof of A0 but only 42% from their own evidence.

Table 3: What agents read and post, rule on vs off. Truth controller, probability matching, $\rho = 0.75$, $q = 6$, $q_c = 12$, $b = 3$, all days.

rule	reads from controller	posts of own evidence	posts of pool facts	own facts held
on	1.34%	78.3%	27.3%	5.81
off	1.50%	76.1%	29.3%	5.67

Caveats. One world (task_003), 24 agents, exact-Bayesian agents; numbers are means over 1,000 episodes per cell. Full analysis, code and every table: [analysis/task003_llm_free/sim1_analysis/](#) (reports Q0–Q6).